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Project report

University *of Ljubljana*
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Translation assistant for legal terminology

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Abstract

For this project we are going to create an AI tranlsation assistant, that will be able to translate between Slovenian and English. We will use the T5 transformer architecture, which is a state-of-the-art model for natural language processing tasks. We will train the model on a large dataset of parallel sentences in Slovenian and English, and we will evaluate its performance on a test set of sentences that it has not seen before. We will also enhance our base model with RAG and compare the performances of the two. The model focuses on legal terminology.

Keywords
translation, Slovenian, English, terminology

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Introduction

Accurate translation in specialized domains such as legal and administrative texts is difficult, especially when using general machine translation tools. While systems like Google Translate or large language models can produce fluent text, they often make mistakes with domain-specific terminology or use inconsistent translations. This is not ideal because in institutional contexts precise and standardized terminology is key for clarity and correctness.

Since the introduction of the Transformer architecture, Transformer models have revolutionized the field of machine translation and have replaced older models, such as CNNs and RNNs that process data word by word, unlike Transformer models that work with the text as a whole. Some recently released large language models such as Llama and Gemma adapt this concept and provide more flexible and context-aware text generation, nonetheless domain-specific translation is still difficult because of a lack of consistent terminology and factual accuracy.

In recent years, new approaches such as Retrieval-Augmented Generation (RAG) have been proposed to improve this. RAG combines information retrieval with text generation: instead of translating only based on the model, it first retrieves relevant documents and then uses them to produce a more accurate and grounded output.

There are also many existing tools and frameworks for language processing, such as NLTK or spaCy, but they are general-purpose and do not focus on domain-specific translation.

For this project, we focus on the institutional and administrative domain and translation from English into Slovenian. The main corpus used is the DGT Translation Memory, which contains officially translated documents from the European Commission. These texts include high-quality legal, administrative and policy-related parallel data in 24 official EU languages, including English and Slovenian, and are suitable for extracting domain-specific terminology and context and improving translation quality.

The goal of this project is to build a translation assistant that produces more accurate and consistent legal translations based on retrieved institutional texts.

Methods

First of our proposed corpora is the DGT Translation Memory, which is a multilingual parallel corpus made out of .tmx documents from the European Commission Agency[1]. For this project we will extract only the English and Slovenian segments. We will supplement this with aligned English-Slovenian texts from EUR-Lex, which includes EU treaties, regulations and directives[2]. For terminology reference we will use the IATE - a bilingual glossary of legal and administrative terms[3].

Implementation

We constructed our dataset from the DGT Translation Memory, which contains parallel texts from European Union institutions. From this corpus, we extracted aligned English–Slovenian sentence pairs and converted them into a structured CSV file format for easier processing. We randomly sampled a subset of 500 sentence pairs, which were used for translation and evaluation during this phase. Additionally, we created a separate test subset of 200 sentence pairs, which will be used later on unseen data.

The dataset consists of parallel sentences, where each English sentence is paired with its corresponding Slovenian translation. The Slovenian sentences serve as the reference (gold standard) for evaluation.

We implemented a baseline machine translation system using a pretrained transformer model. Instead of training a model from scratch, we used the NLLB-200 model (facebook/nllb-200-distilled-600M), which supports multilingual translation including English and Slovenian.

The model and tokenizer were loaded using the HuggingFace Transformers library. We configured the source language as English (eng_Latn) and the target language as Slovenian (slv_Latn) to ensure correct translation direction.

To improve performance and reduce runtime, translations were processed in batches of eight sentences at a time. This batching strategy significantly speeds up inference compared to processing each sentence individually.

RAG

Retrieval-Augmented Generation (RAG) is a technique that connects Large Language Models (LLMs) to external knowledge sources. Instead of relying solely on information learned during training, a RAG system retrieves relevant information from a knowledge base and provides it to the model as additional context during inference. This approach can improve the factual accuracy of generated outputs and reduce hallucinations, where a model produces incorrect information, fabricated citations, or unsupported statements [4, 5].

The retrieval corpus used by the RAG component was created from the training data used for the basic model.

Since semantic retrieval relies on vector representations of text, all English sentences in the knowledge base were converted into embeddings using the multilingual SentenceTransformer model *paraphrase-multilingual-MiniLM-L12-v2*. These embeddings capture the semantic meaning of sentences, allowing semantically similar sentences to be identified even when they differ lexically.

The sentence embeddings were indexed using FAISS (Facebook AI Similarity Search), a library designed for efficient similarity search in high-dimensional vector spaces. Because semantic similarity was measured using cosine similarity, all embedding vectors were normalized to unit length before indexing. After normalization, cosine similarity becomes equivalent to the inner product between vectors, allowing FAISS's IndexFlatIP index to efficiently retrieve the most semantically similar sentences from the knowledge base.

During translation, the input English sentence is first encoded into an embedding using the same SentenceTransformer model. The embedding is then used as a query to search the FAISS index for the most semantically similar sentence in the knowledge base. In our implementation, the single most similar example (`k=1`) is retrieved together with its corresponding Slovenian translation.

The retrieved English–Slovenian sentence pair is inserted into the prompt as a translation example. The FLAN-T5 model then receives both the retrieved example and the new sentence to be translated. This provides the model with contextual guidance based on a similar previously seen translation before generating the final Slovenian output.

Evaluation

For quantitative evaluation, we used the BLEU (Bilingual Evaluation Understudy) metric, implemented via the SacreBLEU library. BLEU measures the similarity between machine-generated translations and reference translations based on overlapping n-grams. We also used the Character n-gram F-score (ChrF) metric, which measures similarity using character-level n-grams rather than word-level n-grams. As a result, it often aligns more closely with human judgment by recognizing partial matches and morphological variations that word-based metrics may overlook.

The baseline model achieved a BLEU score of 36.17 when using sampled data from the NLLB dataframe, which indicates good translation quality. The NLLB dataframe, consists in parallel text, specifically designed for translation, which explains the strong performance of the model. When the more general-purpose FLAN-T5 model is used for the same translation task, the BLEU score drops to 0.12, whereas the RAG-based model achieves a BLEU score of 23.

Because BLEU is generally considered an older metric and can be overly sensitive to exact word matches, we additionally evaluated the baseline and RAG-based models on the FLAN-T5 dataset using only the ChrF metric. The baseline model achieved a ChrF score of 17.52, while the RAG-based model achieved a score of 41.88. These results suggest that the RAG-based approach produces translations that are considerably closer to the reference texts, even when exact word overlap is limited.

As for qualitative analysis, we took a small subset of our test set and manually checked for errors of RAG translations. Our subset consisted of 30 examples and we examined every 30th translation, starting with the translation with index number 0. The possible errors of RAG translation, compared to the Gold standard, were the following:

- **Grammatical error** = the translation consisted of grammatical errors in Slovene language, such as lack of diacritics (č, š, ž) and wrong inclinations, etc.
- **No translation** = the RAG translation was the same as the source sentence
- **Wrong target language** = the RAG translation involved partial or complete switching to another target language

- **Nonsensical output** = severe generation failures where the produced RAG translation was not interpretable
- **No major error** = the translation had no major flaws as/or it was the same as the Gold standard

Here are screenshoted examples of possible errors, taken from our Google Collab's Notebook:

No translation

source	Harrows
gold_translation	brane z zobci
baseline_translation	Harrows, Harrows, Harrows, Harrows, Harrows, H...
rag_translation	Harrows

Wrong target language

source	Residency required for approval.
gold_translation	Za odobritev je potrebno stalno prebivališče.
baseline_translation	Residency required for approval
rag_translation	Residenci requerio a aprovizionaci.

Nonsensical output

source	Single grain whisky and blended grain whisky, ...
gold_translation	single grain whisky in blended grain whisky, v...
baseline_translation	Single grain whisky and blended grain whisky, ...
rag_translation	Srdce srdce a srdce srdce srdce srdce srdce sr...

No major error

source	research, development, demonstration, deployme...
gold_translation	raziskave, razvoj, predstavitev, uporaba in ra...
baseline_translation	research, development, demonstration, deployme...
rag_translation	raziskave, razvoj, predstavitev, uporaba in ra...

Grammatical error

source	Calcium salts of palm fatty acid
gold_translation	kalcijeve soli iz maščobnih kislin palmovega olja
baseline_translation	Calcium salts of palm fatty acid
rag_translation	kalcijeve soli iz maobnih kislin palmovega olja

Out of the 30 manually evaluated examples, 11 were flagged as Nonsensical output, 1 as No translation, 9 as Grammatical error, 6 as No major error and 3 as Wrong target language.

Error category	Count	Percentage
Nonsensical output	11	36.7%
Grammatical error	9	30.0%
No major error	6	20.0%
Wrong target language	3	10.0%
No translation	1	3.3%
Total	30	100.0%

Figure 1. Table 1. Qualitative error analysis of 30 manually evaluated RAG translations.

Overall, the results indicate that the model produces reasonably accurate translations when trained and evaluated on domain-specific translation data, while performance decreases on more general-purpose datasets. However, the ChrF evaluation demonstrates that the RAG-based model retains a substantially higher degree of similarity to the reference translations than the baseline model. There is still room for improvement, particularly in handling specialized legal terminology.

References

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